26900043 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCBIF301 - SYSTEMS & SYNTHETIC BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **5** questions

Each question carries **2** marks

1. List two features of Systems Biology. [K] / [CO1]
2. Explain the basic idea of the genetic paradigm. [U] / [CO2]
3. Describe the relationship between nodes and edges in a biological network. [U] / [CO2]
4. Give example of interactions that can act as edges. [U] / [CO2]
5. Explain the meaning of methodology in Synthetic Biology. [U] / [CO5]
6. Describe the major goals of Synthetic Biology. [U] / [CO5]
7. Which gene encodes GFP? [K] / [CO8]

[2x5 = 10]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

8. List the environmental factors that can affect bacterial image quality. [K] / [CO7]

9. Explain the basic structure of a PPI network. [An] / [CO3]

10. List and explain the medical, environmental and industrial benefits of synthetic biology research. [K] / [CO5]

11. Explain the importance of Synthetic Biology. [U] / [CO5]

12. Evaluate how negative feedback can improve robustness. [An] / [CO6]

13. Explain the use of GFP in biosensor development with a suitable example. [U] / [CO8]

14. What are **biosensors**? Give one environmental application. [K] / [CO5]

15. Analyze a simple regulatory circuit and identify its components. [An] / [CO3]

[5x6 = 30]

Part C

Long Essay

Answer any **3** questions

Each question carries **10** marks

16. Discuss basic concepts of flux analysis in biological systems. [U] / [CO4]

17. Analyze the role of simulation in E-Cell and explain how it helps study cellular behaviour. [An] / [CO6]

18. Explain the functions of **lacZ** and β-galactosidase in the Bacterial Camera Project. [K] / [CO7]

19. Discuss the major ethical, environmental and safety concerns associated with the development and application of synthetic organisms. [K] / [CO5]

20. Interpret the mechanism, components and importance of genetic switches. [U] / [CO2]

[10x3 = 30]